

3) $f(x) = \sqrt{x}$ $a = 4$

$L(x) = f(a) + f'(a)(x-a)$

$f(4) = \sqrt{4} = 2$

$f'(x) = x^{1/2}$
 $f'(x) = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$

$L(x) = 2 + \frac{1}{2\sqrt{4}}(x-4)$

$L(x) = 2 + \frac{x-4}{2\sqrt{4}} = \frac{1}{4}x + 1$

13) a) $y = \tan \sqrt{t}$

$y' = \sec^2(\sqrt{t}) \cdot \frac{1}{2\sqrt{t}} = \frac{\sec^2 \sqrt{t}}{2\sqrt{t}}$

b) $y = \frac{1-v^2}{1+v^2}$

$y' = \frac{(2v)(1+v^2) - (1-v^2)(2v)}{(1+v^2)^2} = \frac{-2v-2v^3-2v+2v^3}{(1+v^2)^2} = \frac{-4v}{(1+v^2)^2}$

77) $y = \sqrt{3+x^2}$

$x = 7$

$dx = -0.7$

a) $\frac{dy}{dx} = \frac{1}{2\sqrt{3+x^2}} \cdot 2x = \frac{2x}{2\sqrt{3+x^2}} = \frac{x}{\sqrt{3+x^2}}$

$dy = \frac{x}{\sqrt{3+x^2}} dx$

b) $dy = \frac{1}{\sqrt{3+7^2}} \cdot -0.7 = -\frac{0.7}{20} \approx -0.035$

27) $y = \sqrt{x-2}$

$x = 3$

$\Delta x = 0.8$

$\Delta y = ?$

$x_1 = 3$

$x_2 = 3.8$

$y_1 = \sqrt{3-2} = 1$

$y_2 = \sqrt{3.8-2} = 1.34$

$\Delta y = y_2 - y_1 = 1.34 - 1 = 0.34$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x-2}}$$

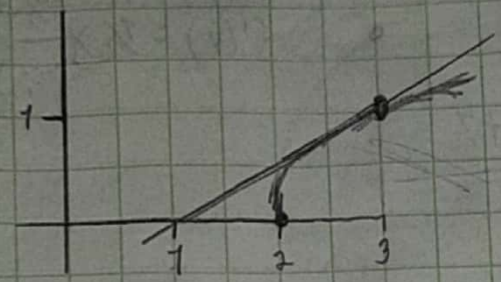
$$dy = \frac{1}{2\sqrt{x-2}} dx$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{3-2}} = \frac{1}{2}$$

$$dy = \frac{1}{2\sqrt{3-2}} \cdot 0.8 = \frac{2}{5} = 0.4$$

$$y - 1 = \frac{1}{2}(x - 3)$$

$$y = \frac{1}{2}x - \frac{1}{2}$$



25) $\sqrt[3]{1001}$

$$f(x) = \sqrt[3]{1000+x}$$

$$a = 1$$

$$L(x) = f(a) + f'(a)(x-a)$$

$$f(a) = \sqrt[3]{1000+1} = 10.0033$$

$$L(x) = 10.0033 + 0.0033(x-1)$$

$$f'(x) = \frac{1}{3}(1000+x)^{-2/3} = \frac{1}{3\sqrt[3]{(1000+x)^2}}$$

$$L(x) = 10.0033 + 0.0033x - 0.0033$$

$$f'(a) = \frac{1}{3\sqrt[3]{(1000+1)^2}} = 0.0033$$

$$L(x) = 10 + 0.0033x$$

$$L(1) = 10 + 0.0033(1)$$

$$L(1) = 10.0033$$

27) $e^{0.7}$

$$f(x) = e^x$$

$$a = 0$$

$$L(x) = f(a) + f'(a)(x-a)$$

$$f'(e) = e^x$$

$$f'(a) = 1$$

$$L(x) = 1 + 1(x-0)$$

$$f(a) = 1$$

$$L(x) = 1 + x$$

$$L(0.7) = 1 + 0.7 = 1.7$$



$$31) \frac{1}{9.98} \approx 0.1002$$

$$f(x) = \frac{1}{x+9} \quad a=7$$

$$L(x) = f(a) + f'(a)(x-a)$$

$$L(x) = 0.1 - 0.01(x-7)$$

$$L(x) = 0.1 - 0.01x + 0.07$$

$$L(x) = 0.1 - 0.01x$$

$$L(0.98) = 0.1 - 0.01(0.98) = 0.1002$$

$$f(x) = (x+9)^{-1}$$

$$f'(x) = -(x+9)^{-2}$$

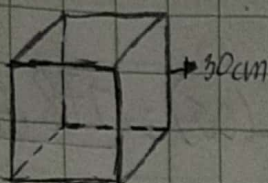
$$f'(x) = \frac{-1}{(x+9)^2}$$

$$f(a) = \frac{1}{1+9} = \frac{1}{10}$$

$$f'(a) = \frac{-1}{(1+9)^2}$$

$$f'(a) = -1/100$$

33)



$$da = 0.1 \text{ cm}$$

$$a) V = a^3$$

$$V = a^3 = 30^3 = 27,000$$

$$dV = 3a^2 da$$

$$\text{Error relative} = \frac{270}{27,000} = 0.01$$

$$\text{" percentual} = 0.01 \cdot 100 = 1\%$$

$$b) A = 6a^2$$

$$\frac{dA}{da} = 12a$$

$$A = 6a^2 = 6(30)^2 = 5,400$$

$$\text{Error relative} = \frac{36}{5,400} = 0.006$$

$$\text{" percentual} = 0.006 \cdot 100 = 0.6\%$$

$$dV = 3a^2 da$$

$$dV = 3(30)^2 (0.1)$$

$$dV = 2,700(0.1) = 270 \text{ cm}^3$$

$$dA = 12a \cdot da$$

$$dA = 12(30)(0.1) = 36 \text{ cm}^2$$

$$35) \text{ circunferencia} = 84 \text{ cm}$$

$$dc = 0.5 \text{ cm}$$

$$a) A = 4\pi r^2$$

$$C = 2\pi r$$

$$\frac{dC}{dr} = 2\pi$$

$$C = 2\pi r$$

$$84 = 2\pi r$$

$$\frac{84}{2\pi} = r = 13.3690$$

$$\frac{dA}{dr} = 8\pi$$

$$dA = 8\pi r dr$$

$$dc = 2\pi dr$$

$$\frac{0.5}{2\pi} = dr = 0.0796$$

$$dA = 8\pi (13.3690)(0.0796) = 26.79$$

$$A = 4\pi (13.3690)^2 = 2245.99 \text{ cm}^2$$

$$\text{Error relative} = \frac{26.79}{2245.99} = 0.0119$$

$$b) V = \frac{4}{3} \pi r^3$$

$$V = \frac{4}{3} \pi \left(\frac{C}{2\pi} \right)^3$$

$$\frac{C}{2\pi} = r$$

$$V = \frac{4}{3} \pi \frac{C^3}{8\pi^3} = \frac{C^3}{6\pi^2}$$

$$\frac{dV}{dC} = \frac{3C^2}{6\pi^2}$$

$$V = \frac{84^2}{6\pi^2} = 7008.91 \text{ cm}^3$$

$$dV = \frac{C^2}{2\pi^2} dC$$

$$\text{Error static} = \frac{778.73}{7008.91} = 0.1111 \quad \Delta V = \frac{84^2}{2\pi^2} (0.5) = 778.73$$